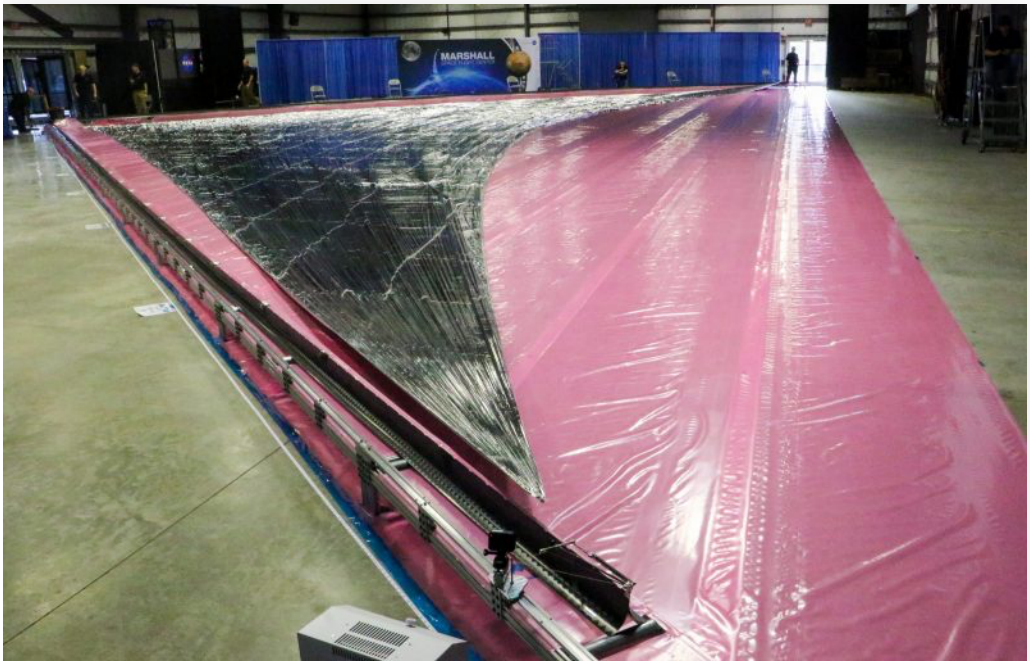
Deploys Largest-Ever Solar Sail at NASA's Marshall Space Flight Center



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Redwire is sail-ebrating the success of its Solar Sail System, developed for NASA Marshall Space Flight Center's (MSFC) Solar Cruiser spacecraft. Redwire's Solar Sail System completed a ground demonstration at NASA MSFC in Huntsville, Alabama, a milestone which propelled the capability to Technology Readiness Level 5 and represents the culmination of various NASA mission reviews, including System Requirement Review & Preliminary Design Review. The team also completed a Redwire Internal Critical design review and Manufacturing Readiness Review of the deployment system. At over 120 feet long by 120 feet wide when fully deployed, Solar Cruiser's solar sail will reach across 18,000 square feet — more than three basketball courts side-by-side!

While traditional spacecraft leverage thrusters and heavy expendable propellants to navigate the cosmos, solar sail technology harnesses the power of sunlight to propel spacecraft forward, enabling long-duration spaceflight to non-traditional orbits. Solar sails could unlock new possibilities, such as advanced solar storm warning systems, enhanced lunar communications coverage, and long-range interstellar missions –– at a much lower cost than conventional systems. For Solar Cruiser, Redwire's Solar Sail System will enable unprecedented observations of the Sun from novel and



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difficult-to-reach vantage points.

As seen in the video below, the ground demonstration included a 4,500 square foot sail (roughly one basketball court), two 100-foot composite booms, and a deployment system. The next step for Redwire's Solar Sail System is a series of tests meant to simulate the harsh environment of launch and space, including vibration testing, thermal vacuum testing and an additional deployment test. Following the successful completion of these tests, Redwire's Solar Sail System will reach Technology Readiness Level 6, which signifies a flight-like design that is ready to go to space.

The Solar Cruiser team, led by NASA MSFC, includes both Redwire and its subcontractor NeXolve, which provided the sail membrane.

Check out this video to learn more about Redwire's Solar Sail System.



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